Subtract, square, then add

Observed: (1, 3, 3, 5) Fitted: (1.2, 2.4, 3.6, 4.8) Mean: 3

A TOTAL / SST

$$t_i = y_i - \bar{y}$$

Observation	i = 1	i = 2	i = 3	i = 4	
1 Difference	−2.0	0.0	0.0	+2.0	3 ADD
2 Square	**4.00**	+ **0.00**	+ **0.00**	+ **4.00**	→ SST = 8.00

B EXPLAINED / SSR

$$g_i = \hat{y}_i - \bar{y}$$

Observation	i = 1	i = 2	i = 3	i = 4	
1 Difference	−1.8	−0.6	+0.6	+1.8	3 ADD
2 Square	**3.24**	+ **0.36**	+ **0.36**	+ **3.24**	→ SSR = 7.20

C RESIDUAL / SSE

$$e_i = y_i - \hat{y}_i$$

Observation	i = 1	i = 2	i = 3	i = 4	
1 Difference	−0.2	+0.6	−0.6	+0.2	3 ADD
2 Square	**0.04**	+ **0.36**	+ **0.36**	+ **0.04**	→ SSE = 0.80

The minus signs belong to differences. Every square and every SS is nonnegative.